MOHAMMADAMIN DARIJANI

Research Fellow | Medical Biotechnology

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EDUCATION

University of Piemonte Orientale Novara, Italy / **2023 - 2025**

- **MSc** in **Medical Biotechnology**, Department of Health Sciences
  - **Major:** Cell Therapy, Tissue Engineering and Regenerative Medicine
  - ***Dissertation:*** *Cytocompatibility and Antibacterial Evaluation of Copper and Zinc Co-Doped Plasma Electrolytic Oxidation (PEO)-Treated Titanium Surfaces for Dental Implant Abutment: An In vitro Study*

Kerman University of Medical Sciences Kerman, Iran / **2015 - 2022**

- **Doctorate** Degree in **Dentistry**, Faculty of Dentistry
  - ***Dissertation:*** *Evaluation of Cleaning Efficacy, Instrumentation Time, and Dentin Removal of Manual and Rotary Instrumentation Techniques in Primary Molars: An In vitro Study*

RESEARCH EXPERIENCE

CAAD (Translational Research Centre for Autoimmune and Allergic Diseases) Novara, Italy / **2025 - Present**

Research Fellow, INNOVATION Laboratory (Tissue Engineering & Biomaterials Evaluation)

- Contribute to research projects and laboratory testing services.
- **CeramTec GmbH:** Assessed cell viability, adhesion, proliferation and osteogenic potential in response to polyaryletherketone (PAEK) materials.
  - ***Experimental models:*** *2D cultures of human bone marrow-derived mesenchymal stem cells (hBM-MSCs) and human osteosarcoma cells (U2OS).*
- **Tiss'You srl:** Evaluated cellular responses to low-molecular-weight hydrolyzed collagen peptides, assessing regenerative potential, anti-scarring activity, cytoskeletal organization and cellular mechanobiology.
  - ***Experimental models:*** *2D cultures of human dermal fibroblasts (hTERT-HDFa) and human keratinocytes (HaCaT); 3D reconstructed human epidermis (RHE).*
- **Sanitpharma srl:** Analyzed particle size and elemental composition to assess size-based separation performance and potential particle release from processing filters.

University of Piemonte Orientale Novara, Italy / **2024 - 2025**

Research Intern, Regenerative Anatomy Laboratory

- Conducted physicochemical, biological and antibacterial evaluations of titanium dental implant surfaces modified with copper- and zinc-doped PEO coatings.
- **Physicochemical characterization:** coating topography, surface elemental composition, wettability and surface energy.
- **Biological evaluation:** cytocompatibility, cell morphology, focal adhesions, cytoskeletal organization, inflammatory responses, extracellular collagen deposition and matrix mineralization.
  - ***Experimental models****: 2D cultures of mouse pre-osteoblasts (MC3T3-E1), normal human dermal fibroblasts (NHDF-Ad), human gingival fibroblasts (HGF-1) and human osteoblasts (HOB).*
- **Antibacterial evaluation**: bacterial metabolic activity, viable counts, membrane-integrity and biofilm morphology.
  - ***Experimental models****: sessile cultures of S. epidermidis, S. aureus and A. actinomycetemcomitans.*

University of Oslo Oslo, Norway / **2024**

Erasmus+ Research Trainee, Department of Biomaterials

- Investigated multifunctional tannic acid nanocoating for bone-anchored implant surfaces with reduced infection risk.

- **Surface modification:** prepared tannic acid-coated titanium substrates.
- **Antibacterial evaluation:** activity of UV-inactivated *L. crispatus* biofilms against *S. epidermidis* on saliva-conditioned, tannic acid-coated titanium surfaces.

Lab SKILLS

- **Cell culture**: 2D and 3D mammalian cell culture; primary, immortalized, tumor-derived and stem cells.
- **Microbiology**: planktonic and sessile cultures, biofilm models and antibacterial testing.
- **Cell biology:** assessment of cell viability, metabolic activity, cytotoxicity, adhesion, proliferation, differentiation and maturation.
- **Molecular biology:** gene expression analysis by RT-qPCR (RNA isolation, cDNA synthesis and qPCR)
- **Staining:** histological, histochemical and immunofluorescence staining.
- **Microscopy**: light, fluorescence, confocal and scanning electron microscopy
- **Surface characterization:** optical profilometry, energy-dispersive X-ray spectroscopy (EDS), contact-angle and surface-energy analysis.

PUBLICATIONS

Cu/Zn-doped PEO titanium with waveform-tunable bioactivity for dental implants. E Canciani, M Pavarini, **M Darijani**, Z Elyasi, M Moscatelli, A Scalia, E De Giglio, H Tiainen, A Rogala, R Chiesa, F Boccafoschi. *Surfaces and Interfaces*. 2026;101:111016. https://doi.org/10.1016/j.surfin.2026.111016.

Oral health-related quality of life among children aged 11-14 years old with and without parental care in south-east of Iran. H Kamyabi , **M Darijani** , S Amanpour , V Yazdi-Feyzabadi , R Fereidooni , S Fardisi. Pesquisa Brasileira em Odontopediatria e Clínica Integrada. 2023;23:e220103. https://doi.org/10.1590/pboci.2023.069.

CONFERENCES

Advancing dental implant technology: Bioactive surface modification for optimized interactions with the oral environment. E Canciani, M Pavarini, **M Darijani**, A Scalia, Z Elyasi, M Moscatelli, E De Giglio, R Chiesa, F Boccafoschi. Co-authored oral presentation at the 34th Annual Conference of the European Society for Biomaterials (ESB 2025), Torino, Italy, September 2025.

PROFESSIONAL EXPERIENCE

Cochrane Part-time / Remote / **2022 - 2023**

Office Assistant, Cochrane Iran Associate Center

FANAP's Advanced Technologies Lab (ZLab) Contract / Remote / **2022 - 2023**

Data Annotator & Clinical Expert

Kerman University of Medical Sciences Full-Time / Kerman, Iran / **2020 - 2022**

Dental Intern, Dental Faculty Hospital

Ghalamchi Educational Foundation Contract / Kerman, Iran / **2016 - 2018**

Teacher & Educational Consultant

TRAININGS

- "Python, Intro to Programming and Machine Learning, Pandas, Intro to AI Ethics, Data Visualization, Intermediate Machine Learning" Kaggle / **2024**
- "Data Science Math Skills" Duke University / **2023**
- "The Data Scientist's Toolbox" from "Data Science: R Specialization" Johns Hopkins University / **2022**

- "SPSS 1, 2" & "Systematic Review and Meta-Analysis" SSRC / **2021**
- "An Introduction to Meta-Analysis Software; Stata, CMA, RevMan, R" Cochrane Iran / **2021**
- "SPSS Statistical Analysis Software" 5th International Congress on BioMedicine / **2021**
- "Cell Culture" 5th International Congress on BioMedicine / **2021**
- "Bioinformatics and Data Analysis" 5th International Congress on BioMedicine / **2021**
- "Comprehensive Course Rapid Research Education (RRE)" Kerman University of Medical Sciences / **2017**

HONORS & AWARDS

- Research fellowship in innovative dental materials from UniUPO, Italy **2025**
- Erasmus funding from the EU and Piemonte Region, Italy **2024**
- Full merit-based DSU-EDISU Piemonte scholarship, Italy **2023**
- State-funded bursary from Kerman University of Medical Sciences, Iran **2015**

CONTRIBUTIONS & MEMBERSHIPS

- Contribution / UPO Anatomy Talents" - SIMNOVA Simulation & Innovative Teaching Center, Italy **2024 & 2025**
- Contribution / Regenerative Medicine Academy" - NODES Spoke 5 event, Italy **2024 & 2025**
- Licensed Member / Medical Council of I.R. IRAN, DDS / MC. No: D-203558 **2022**
- Member / Faculty Student Research Committee of Kerman University of Medical Sciences **2021**
- Member / Rezvan Kerman Dental Relief Charity - Community Dentist & Health provider **2021**
- Member / Faculty Student Union Council of Kerman University of Medical Sciences **2016**

References available upon request.